~~Review of~~ Functional analysis concepts

Disclaimer

These are concepts necessary to understand many aspects of the theory of PDEs as well of the numerical methods of the PDEs.
They are a bit technical, I don't expect you to remember everything, but to understand why we need to introduce some concepts, in particular, when we will use them in the future lectures.

Source: Quarteroni

Divergence theorem Wiki

Recall: **fundamental theorem of calculus** in 1D: if $f \in C^1([a,b])$, then

$$\int_a^b f'(x)\mathrm{d}x = f(b) - f(a).$$

Generalization to higher dimensions of fundamental theorem of calculus

Let $\Omega \subset \mathbb{R}^d$ be a compact subset with a piecewise smooth boundary $\partial\Omega$, and let $u : \Omega \to \mathbb{R}$ be a continuously differentiable function on an open neighborhood O of $\bar{\Omega} \subset O$, i.e. $u \in C^1(O)$. Then, for each $i = 1, \ldots, d$,

$$\int_\Omega \partial_{x_i} u(x)\mathrm{d}x = \int_{\partial\Omega} u(x) n_i(x)\mathrm{d}S,$$

where $n(x) = (n_1(x), \ldots, n_d(x))$ is the outward unit normal vector to $\partial\Omega$ at x, and $\mathrm{d}S$ is the surface measure on $\partial\Omega$.

Divergence theorem (special case of above)

Let $\Omega \subset \mathbb{R}^d$ be a compact subset with a piecewise smooth boundary $\partial\Omega$, and let $\mathbf{u} : \Omega \to \mathbb{R}^d$ be a continuously differentiable vector field function on an open neighborhood O of $\bar{\Omega} \subset O$, i.e. $u_i \in C^1(O)$ for all $i = 1, \ldots, d$. Then, for each $i = 1, \ldots, d$,

$$\int_\Omega \operatorname{div}(\mathbf{u})\mathrm{d}x = \int_\Omega \sum_{i=1}^d \partial_{x_i} u_i(x)\mathrm{d}x = \int_{\partial\Omega} \sum_{i=1}^d u_i(x) n_i(x)\mathrm{d}S = \int_{\partial\Omega} \mathbf{u}(x) \cdot \mathbf{n}(x)\mathrm{d}S,$$

where $n(x) = (n_1(x), \ldots, n_d(x))$ is the outward unit normal vector to $\partial\Omega$ at x, and $\mathrm{d}S$ is the surface measure on $\partial\Omega$.

Integration by parts in high dimension

Let $f \in C^1(\Omega)$ and $\mathbf{u} \in (C^1(\Omega))^d$, then

$$\int_{\partial\Omega} f\mathbf{u} \cdot \mathbf{n}\mathrm{d}S = \int_\Omega \operatorname{div}(f\mathbf{u}) = \int_\Omega f\operatorname{div}(\mathbf{u})\mathrm{d}x + \int_\Omega \nabla f \cdot \mathbf{u}\mathrm{d}x.$$

Functional spaces

We are used to think about vector spaces as $\mathbb{R}^n$ or $\mathbb{C}^n$. However, we can also think about spaces of functions as vector spaces. For example, the space of continuous functions on an interval $[a, b]$ is a vector space, since the sum of two continuous functions is continuous and the scalar multiplication of a continuous function is also continuous.
We will call them **functional spaces** as their elements are functions.

Examples of (normed) functional spaces

- $C(\Omega)$: the space of continuous functions on Ω, with the supremum norm $||f||_\infty = \sup_{x\in\Omega} |f(x)|$.
- $C^k(\Omega)$: the space of functions on Ω that are k-times continuously differentiable, with the norm $||f||_{C^k} = \sum_{p_1+\cdots+p_d\leq k} ||\partial_{x_1^{p_1}\ldots x_d^{p_d}} f||_\infty$.
- $C^\infty(\Omega)$: the space of functions on Ω that are infinitely differentiable.
- $L^2(\Omega) := \{f : \Omega \to \mathbb{R} : \int_\Omega |f(x)|^2 dx < \infty\}$: the space of square-integrable functions on Ω, with the norm $||f||_{L^2} = \sqrt{\int_\Omega |f(x)|^2 dx}$.

Keep these examples in mind, we will see them again and again!

Exercise/Homework

Show that $||f||_{L^2} = \sqrt{\int_\Omega |f(x)|^2 dx}$ is a norm in $L^2(\Omega) := \{f : \Omega \to \mathbb{R} : \int_\Omega |f(x)|^2 dx < \infty\}$: the space of square-integrable functions on Ω.

Recall: a norm is a function $||\cdot|| : V \to \mathbb{R}$ that satisfies the following properties for all $u, v, w \in V$ and $\alpha \in \mathbb{R}$:

1. $||u|| \geq 0$ and $||u|| = 0$ if and only if $u = 0$ (positive definiteness);
2. $||\alpha u|| = |\alpha| ||u||$ (homogeneity);
3. $||u + v|| \leq ||u|| + ||v||$ (triangle inequality).

Spoiler: there are infinitely many functions in L^2 that have $||\cdot||_{L^2} = 0$. How to fix this?
We change the definition of L^2 as the space of equivalence classes of functions that are equal almost everywhere, i.e.,

$$L^2(\Omega) := \frac{\{f : \Omega \to \mathbb{R} : \int_\Omega |f(x)|^2 dx < \infty\}}{\sim}$$

where $f \sim g$ if $f(x) = g(x)$ for almost every $x \in \Omega$.

Linear functionals

Given a functional space V, a **linear functional** is a map $L : V \to \mathbb{R}$ that satisfies linearity: $L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$ for all $u, v \in V$ and scalars $\alpha, \beta \in \mathbb{R}$.

Examples

- The evaluation functional $L_{x_0} : C(\Omega) \to \mathbb{R}$ defined by $L_{x_0}(u) := u(x_0)$ for a fixed $x_0 \in \Omega$ is a linear functional;
- The integral functional $L : L^2(\Omega) \to \mathbb{R}$ defined by $L(u) := \int_\Omega u(x)dx$ is a linear functional;
- The bilinear form $B : L^2(\Omega) \times L^2(\Omega) \to \mathbb{R}$ defined by $B(u, v) := \int_\Omega u(x)v(x)dx$ is a bilinear functional.

Linear functionals properties

Boundedness and Continuity

A functional L is bounded if there exists a constant C such that $|L(u)| \leq C||u||_V$ for all $u \in V$. If V is a Banach space (normed and complete), then a linear bounded functional is also continuous.

Example of unbounded functional

The "evaluation in 0 of the derivative" functional $L : C^\infty([0,1]) \to \mathbb{R}$ defined by $L(u) := u'(0)$ is not bounded with respect to the supremum norm, since for any $C > 0$, we can find a function u such that $||u||_\infty = 1$ but $|L(u)| = |u'(0)| \to \infty$, e.g.

$$u_n(x) = \sin(n\,x) \implies ||u_n||_\infty = 1, \quad L(u_n) = u_n'(0) = n \to \infty.$$

Dual Space

The dual space $V^* = V'$ is the space of all bounded linear functionals on V.

$$V^* := \{F : V \to \mathbb{R} : \ F \text{ is linear and bounded}\}.$$

Norm

The norm of a functional $L \in V^*$ is defined as
$||L||_{V^*} = \sup_{||u||_V \leq 1} |L(u)| = \sup_{||u||_V \neq 0} \frac{|L(u)|}{||u||_V}$.

Homework

Show that the two formulae above are equivalent $\sup_{||u||_V \leq 1} |L(u)| = \sup_{||u||_V \neq 0} \frac{|L(u)|}{||u||_V}$.

Hilbert Space

A Hilbert space H is a real or complex inner product space that is also a *complete* metric space with respect to the distance function induced by the inner product.

The inner product is a bilinear function $(\cdot,\cdot)_H : V \times V \to \mathbb{R}$ that is symmetric and positive definite. The induced norm is $||u||_H := \sqrt{(u,u)_H}$.

Definition (Complete space)

A complete space is a space where every Cauchy sequence converges to a limit within the same space, i.e., for every $\{u_n\}_{n\in\mathbb{N}} \subset V$ such that

$$\lim_{m,n\to\infty} ||u_m - u_n||_V = 0,$$

there exists $u \in V$ such that $\lim_{n\to\infty} ||u_n - u||_V = 0$.

Examples

- $L^2(\Omega)$ is a Hilbert space with the inner product defined as $(f,g)_{L^2} = \int_\Omega f(x)g(x)dx$.
- $C^0(\Omega)$ with the supremum norm is not a Hilbert space, since it is not complete with respect to the supremum norm. For example, the sequence of functions $f_n(x) = x^n$ converges pointwise to the function $f(x) = 0$ for $x \in [0,1)$ and $f(1) = 1$, which is not continuous, so it does not belong to $C^0(\Omega)$.

Riesz Representative

The **Riesz representation theorem** states that for every bounded linear functional L on a **Hilbert space** H, there exists a unique element $v_L \in H$ such that

$$L(u) = (u, v_L)_H$$

for all $u \in H$. Moreover, $||L||_{H^*} = ||v_L||_H$.
Conversely, for every element $u \in H$ there exists a linear and bounded functional L_u such that

$$L_u(v) = (u, v)_H \text{ for every } v \in H.$$

Moreover, $||L_u||_{H^*} = ||u||_H$.

Hence, there is a bijection between H and H^*.

Bilinear form definitions

Given V a normed functional space, a bilinear form a is a function that maps every two elements of V to a scalar

$$a : V \times V \to \mathbb{R}.$$

A form

- is **bilinear** if
  - $a(\lambda u + \mu w, v) = \lambda a(u, v) + \mu a(w, v)$ for every $\lambda, \mu \in \mathbb{R}$ and every $v, w, u \in V$, and
  - $a(u, \lambda v + \mu w) = \lambda a(u, v) + \mu a(u, w)$ for every $\lambda, \mu \in \mathbb{R}$ and every $v, w, u \in V$;
- is **continuous** if there exists an $M > 0$ such that

$$a(u, v) \leq M \|u\|_V \|v\|_V \text{ for every } v, u \in V;$$

- is **symmetric** if $a(u, v) = a(v, u)$ for every $u, v \in V$;
- is **positive** if $a(v, v) > 0$ for all $v \in V$ with $v \neq 0$;
- is **coercive** if there exists $\alpha > 0$ such that $a(v, v) > \alpha \|v\|_V^2$ for all $v \in V$.

Homework

- $a(u,v) := \int_{\mathbb{R}} \sin(x)^2\, u(x)v(x)\,\mathrm{d}x$ is bilinear, continuous, symmetric, positive but not coercive on $L^2(\mathbb{R})$
- $a(u,v) := \int_{\mathbb{R}} u(x)v(x)\,\mathrm{d}x$ is bilinear, continuous, symmetric, positive and coercive on $L^2(\mathbb{R})$
- $a(u,v) := \int_{\mathbb{R}} \sin(x)\, u(x)v(x)\,\mathrm{d}x$ is bilinear, continuous, symmetric, not positive nor coercive on $L^2(\mathbb{R})$
- $a(u,v) := \int_{\mathbb{R}} \sin(u(x))\, v(x)\,\mathrm{d}x$ is continuous, is not bilinear, not symmetric, not positive nor coercive on $L^2(\mathbb{R})$.

Distributions

Let $\Omega \subset \mathbb{R}^d$ be an open set and $f : \Omega \to \mathbb{R}$ a function.

Support of a Function

The support of a function f, denoted by $\mathrm{supp}(f)$, is the closure of the set where f is non-zero.

$$\mathrm{supp}(f) := \overline{\{x \in \Omega : f(x) \neq 0\}}.$$

Compact Support

A function has compact support if its support is a compact set (compact = closed and bounded).

C^∞ Compact Support Functions

A function is in $\mathcal{D}(\Omega) := C_c^\infty(\Omega)$ if it is infinitely differentiable and has compact support in Ω.

Convergence in $\mathcal{D}(\Omega)$

A sequence of functions $\{f_n\}$ in $\mathcal{D}(\Omega)$ converges to f in $C_c^\infty(\Omega)$ if

- exists a fixed compact set K that contains all supports of f_n
- all derivatives of f_n converge *uniformly* to the corresponding derivatives of f, i.e. $\partial_{x_1^{p_1}\dots x_d^{p_d}} f_n \to \partial_{x_1^{p_1}\dots x_d^{p_d}} f$ for all $p_1, \dots, p_d$.

Distributions

A **distribution** is a linear functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$ that is continuous, i.e.,

$$\lim_{k\to\infty} T(\varphi_k) = T(\varphi),$$

for all $\varphi_k \to_{\mathcal{D}} \varphi \in \mathcal{D}$.
Hence, the distribution space $\mathcal{D}^*(\Omega)$ is the dual of $\mathcal{D}(\Omega)$.

Notation for distribution $T \in \mathcal{D}^*(\Omega)$ applied to a function $f \in \mathcal{D}(\Omega)$: $T(f) = \langle T, f\rangle$.

Example Dirac Delta

The Dirac delta distribution δ_a with $a \in \Omega$ a point, is defined by $\delta_a(\phi) = \phi(a)$ for all $\phi \in \mathcal{D}(\Omega)$. It is a distribution that "picks out" the value of a function at a point.
The Dirac delta is not a functional on the $L^2(\mathbb{R})$ space, since it is not bounded with respect to the L^2 norm.
However, it is a distribution in $\mathcal{D}^*(\Omega)$.

Convergence in $\mathcal{D}^*(\Omega)$

A sequence of distributions T_n converges in $\mathcal{D}^*(\Omega)$ to $T \in \mathcal{D}^*(\Omega)$ if

$$\lim_{n\to\infty} T_n(\varphi) = T(\varphi), \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

$L^2(\Omega)$ squared summable functions

$$L^2(\Omega) := \frac{\left\{ f : \Omega \to \mathbb{R} \text{ such that } \int_\Omega f(x)^2 \mathrm{d}x < \infty \right\}}{\sim},$$

$f \sim g$ if $f(x) = g(x)$ for almost every $x \in \Omega$.

1. $L^2(\Omega)$ is a Hilbert space with scalar product $(f, g) := \int_\Omega f(x)g(x)\mathrm{d}x$.
2. The $L^2(\Omega)$ norm is define through the inner product as $\|f\|_{L^2(\Omega)} := \sqrt{\int_\Omega f(x)^2 \mathrm{d}x}$.
3. To every function $f \in L^2(\Omega)$ is associated a distribution $T_f \in \mathcal{D}^*(\Omega)$ defined by
$$T_f(\varphi) := \int_\Omega f(x)\varphi(x)\mathrm{d}x, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$
4. $\mathcal{D}(\Omega)$ is dense in $L^2(\Omega)$, i.e., for every $f \in L^2(\Omega)$ there exists a sequence of $\varphi_k \in \mathcal{D}(\Omega)$ such that
$$\|\varphi_k - f\|_{L^2(\Omega)} \to 0.$$
5. $\mathcal{D}(\Omega) \subset L^2(\Omega) \Longrightarrow (L^2(\Omega))^* = L^2(\Omega) \subset \mathcal{D}^*(\Omega)$.

Example: convergence to Dirac Delta

Let $\chi_{[a,b]}$ be the characteristic function on the interval $[a,b] \subset \mathbb{R}$ defined as

$$\chi_{[a,b]}(x) = \begin{cases} 0 & \text{if } x \notin [a,b], \\ 1 & \text{if } x \in [a,b]. \end{cases}$$

Let us build the sequence of functions in $L^2(\mathbb{R})$ $f_n(x) := \frac{n}{2}\chi_{[-1/n,1/n]}(x)$. Clearly, we have that

1. $\int_{\mathbb{R}} f_n(x)\mathrm{d}x = 1$
2. $T_{f_n}(\varphi) = \int_{\mathbb{R}} f_n(x)\varphi(x)\mathrm{d}x = \frac{n}{2}\int_{-1/n}^{1/n} \varphi(x)\mathrm{d}x = \frac{n}{2}(\Phi(1/n) - \Phi(-1/n))$ where $\frac{d}{dx}\Phi(x) = \varphi(x)$.
3. Let $h_n = 1/n$, $T_{f_n}(\varphi) = \frac{\Phi(h_n)-\Phi(-h_n)}{2h_n}$
4. $\lim_{n\to\infty} T_{f_n}(\varphi) = \lim_{n\to\infty} \frac{\Phi(h_n)-\Phi(-h_n)}{2h_n} = \frac{d}{dx}\Phi(0) = \varphi(0)$.
5. $T_{f_n}(\varphi) \to \varphi(0) = \delta_0(\varphi)$.
6. δ_0 is a distribution in $\mathcal{D}^*(\mathbb{R})$ but not a function in $L^2(\mathbb{R})$.

Definition: Derivation in distributional sense

Let $T \in \mathcal{D}^*(\Omega)$, with $\Omega \subset \mathbb{R}^d$ open. We can define the derivative of T using the integration by parts.

$$\partial_{x_i} T(\varphi) = \langle \partial_{x_i} T, \varphi \rangle := -\langle T, \partial_{x_i} \varphi \rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

If T is a T_f with $f \in \mathcal{C}^1(\Omega)$, it is clearly the classical derivative. Let's see in 1D with $\Omega = [a, b]$.

$$\partial_x T_f(\varphi) = \langle \partial_x T_f, \varphi \rangle = \int_a^b \partial_x f(x) \varphi(x) \mathrm{d}x = \underbrace{[f(x)\varphi(x)]_a^b}_{=0} - \int_a^b f(x) \partial_x \varphi(x) \mathrm{d}x, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

But the definition holds for every distribution, even for discontinuous functions, for the Dirac delta, etc. !!!

Higher derivatives

$$\left\langle \frac{\partial^{p_1 + \cdots + p_d} T}{\partial x_1^{p_1} \ldots \partial x_d^{p_d}}, \varphi \right\rangle := (-1)^{p_1 + \cdots + p_d} \left\langle T, \frac{\partial^{p_1 + \cdots + p_d} \varphi}{\partial x_1^{p_1} \ldots \partial x_d^{p_d}} \right\rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

Example: Derivative of Heaviside function

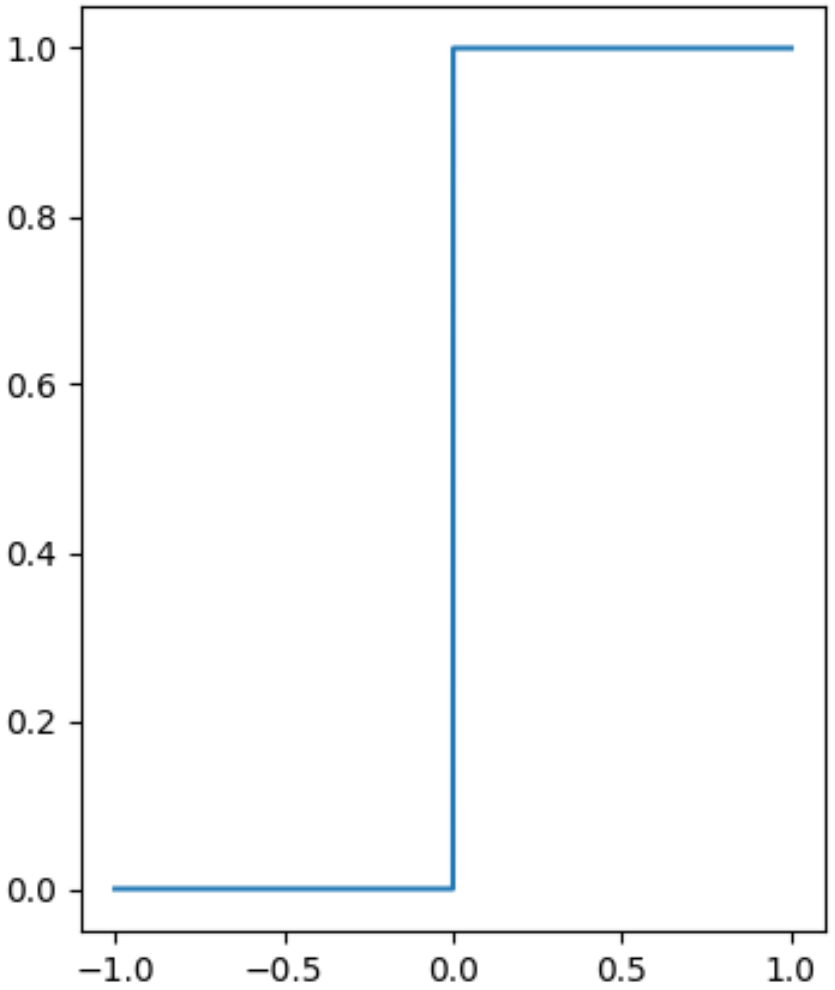
$$R(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$$

- $R \in L^2((-1,1))$
- $R \notin C((-1,1))$
- $T_R \in \mathcal{D}^*((-1,1))$

$$\langle \partial_x T_R, \varphi \rangle = -\int_{-1}^{1} R(x) \partial_x \varphi(x) \mathrm{d}x$$

$$= -\int_{0}^{1} \partial_x \varphi(x) \mathrm{d}x = -[\varphi]_0^1 = \varphi(0)$$

$$\implies \partial_x R = \delta_0.$$



Homework

- Compute the second derivative $\partial_{xx} T_R$ of the Heaviside function in the distributional sense
  $R(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$
- Compute the derivative of the function $f(x) = |x|$ in the distributional sense and checks that it coincides with the classical derivative where it is differentiable.
- Define the multiplication between a function $f \in \mathcal{C}^\infty$ and a distribution T as $f \cdot T(\varphi) := T(f \cdot \varphi)$. Prove that for $f(x) = x$, we have that $f \cdot \delta_0' = -\delta_0$ in distributional sense.

Sobolev Spaces

As we have seen $(L^2(\Omega))^* = L^2(\Omega) \subset \mathcal{D}^*(\Omega)$. This does not imply that their distributional derivatives are still in L^2. The Heaviside function is in L^2 but its derivative it's not.

We need to introduce other spaces to distinguish functions!

Sobolev spaces

Let $\Omega \subset \mathbb{R}^d$ and $k \in \mathbb{N}_0$. We define the Sobolev space of order k on Ω the space of the functions in $L^2(\Omega)$ with distributional derivatives up to order k in $L^2(\Omega)$.

$$H^k(\Omega) := \{f \in L^2(\Omega) : \partial_{x_1^{p_1} \dots x_d^{p_d}} f \in L^2(\Omega), \text{ for all } p_1, \dots, p_d : p_1 + \dots + p_d \leq k\}.$$

- $H^{k+1}(\Omega) \subset H^k(\Omega)$
- $L^2(\Omega) = H^0(\Omega)$
- Heaviside $R \in H^0((-1,1))$, but $R \notin H^1((-1,1))$

Examples

- Example of $H^\infty(\Omega)$ but not $C(\Omega)$

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 0, \\ 3 & \text{if } x = 0. \end{cases}$$

- Example of $H^1(\Omega)$ but not $H^2(\Omega)$

$$f(x) = \begin{cases} x & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases} \qquad f'(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

Norms and inner products of Sobolev spaces

- Sobolev spaces $H^k(\Omega)$ are Hilbert space with respect to the following scalar product

$$(f,g)_k = (f,g)_{H^k(\Omega)} := \sum_{p_1+\cdots+p_d\le k} \int_\Omega \partial_{x_1^{p_1}\ldots x_d^{p_d}} f \cdot \partial_{x_1^{p_1}\ldots x_d^{p_d}} g \,\mathrm{d}x,$$

  with the norms

$$\|f\|_k = \|f\|_{H^k(\Omega)} := \sqrt{\sum_{p_1+\cdots+p_d\le k} \int_\Omega (\partial_{x_1^{p_1}\ldots x_d^{p_d}} f)^2 \,\mathrm{d}x},$$

- Seminorms

$$|f|_k = |f|_{H^k(\Omega)} := \sqrt{\sum_{p_1+\cdots+p_d= k} \int_\Omega (\partial_{x_1^{p_1}\ldots x_d^{p_d}} f)^2 \,\mathrm{d}x},$$

- $\|f\|_k = \sqrt{\sum_{m=0}^k |f|_m^2}$

Examples for $k = 1$

$$(f, g)_1 = (f, g)_{H^1(\Omega)} = \int_\Omega f(x)\, g(x)\, \mathrm{d}x + \int_\Omega f'(x)\, g'(x)\, \mathrm{d}x$$

$$\|f\|_1 = \sqrt{\int_\Omega f^2(x)\, \mathrm{d}x + \int_\Omega (f'(x))^2\, \mathrm{d}x} = \sqrt{\|f\|^2_{L^2(\Omega)} + \|f'\|^2_{L^2(\Omega)}}$$

$$|f|_1 = \sqrt{\int_\Omega (f'(x))^2\, \mathrm{d}x} = \|f'\|_{L^2(\Omega)}$$

Boundary for bounded domains

Property

If $\Omega \subset \mathbb{R}^d$ is open with a *smooth enough* boundary, then $H^k(\Omega) \subset C^m(\bar{\Omega})$ if $m < k - \frac{d}{2}$.

Careful, in this case we mean that there is a representative of the function in H^k such that it also belongs to C^m. In the previous example where

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 0, \\ 3 & \text{if } x = 0. \end{cases}$$

there exists a continuous representative of this function $f(x) = x^2$ which is the same function in $L^2(\Omega)$.

Examples

$$d = 1, \qquad m < k - \frac{1}{2}, \qquad H^1((-1,1)) \subset C^0([-1,1]).$$

$$d = 2, \qquad m < k - \frac{1}{2}, \qquad H^1((-1,1)) \not\subset C^0([-1,1]), \qquad H^2((-1,1)) \subset C^0([-1,1]).$$

$H_0^1(\Omega)$

Let Ω be a bounded domain. We denote with $H_0^1(\Omega)$ the closure of $\mathcal{D}(\Omega)$ in $H^1(\Omega)$. (*morally zero on the boundary*)

Theorem: Poincarè inequality

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with a Lipschitz boundary. There exists a constant $C = C(\Omega) > 0$ such that for all $u \in H_0^1(\Omega)$,

$$\|u\|_{L^2(\Omega)} \leq C\|\nabla u\|_{L^2(\Omega)} = C|u|_1.$$

Proof

Since $\Omega \subset \mathbb{R}^d$ is bounded there exists a ball $S_R = \{x : |x - x_0| < R\}$ that contains Ω. Since, $\mathcal{D}(\Omega)$ is dense in $H_0^1(\Omega)$, we can prove the inequality for $u \in \mathcal{D}(\Omega)$ and pass to the limit to get it for H_0^1. Notice that $\operatorname{div}(x - x_0) = d$. So, using Cauchy-Schwarz,

$$\|u\|_{L^2(\Omega)}^2 = d^{-1}\int_\Omega d\,|u(x)|^2 \mathrm{d}x = d^{-1}\int_\Omega \operatorname{div}(x - x_0)|u(x)|^2 \mathrm{d}x = -d^{-1}\int_\Omega (x - x_0)\nabla(|u(x)|^2)\mathrm{d}x =$$

$$-2d^{-1}\int_\Omega (x - x_0)u(x)\nabla(u(x))\mathrm{d}x \leq 2d^{-1}\|x - x_0\|_\infty \|u\|_{L^2(\Omega)}\|\nabla u\|_{L^2(\Omega)} = 2d^{-1}R\|u\|_{L^2(\Omega)}|u|_1.$$

Proposition

On $H_0^1(\Omega)$ the seminorm $|\cdot|_1$ is actually a norm and it is equivalent to $\|\cdot\|_1$.

Proof

$$\|u\|_1^2 = |u|_1^2 + \|u\|_{L^2}^2 \leq (1 + C^2)|u|_1^2.$$

On the other hand

$$|u|_1^2 \leq |u|_1^2 + \|u\|_{L^2}^2 = \|u\|_1^2.$$